

Finite Element Methods

Theory, Geometry, and Computation

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February 9, 2026

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Chapter 1

Calculus of Variations

The calculus of variations originated in the late 17th century from problems seeking extrema of functionals rather than functions. A seminal example is the *brachistochrone problem*, posed by Johann Bernoulli in 1696, which asks for the curve of fastest descent under gravity – we will fully solve this problem as an exercise.

The foundations of the field were established in the 18th century, most notably by Euler [1] who derived the first general necessary condition for optimality in the 1740s, now known as the Euler–Lagrange equation. These results were systematized and extended by Lagrange in [3], where the variational approach became a unifying principle in mechanics.

In the 19th century, the variational framework was further refined and generalized, notably through the work of Hamilton [2], who introduced an alternative formulation based on action principles, now fundamental in classical mechanics, optics, and modern field theories. Today, the calculus of variations provides the mathematical foundation for many areas of applied mathematics, including partial differential equations, continuum mechanics, and finite element methods.

One of the most classical and visually striking applications of the calculus of variations is the determination of *minimal surfaces* — surfaces that locally minimize area subject to given boundary constraints. Physically, this problem is exemplified by dipping a wire frame into soapy water: the resulting soap film naturally adopts a shape that minimizes its surface area under the constraint of the fixed boundary provided by the frame. Mathematically, such a surface S can be characterized as a critical point of the area functional

$$\mathcal{A}(u) = \iint_{\Omega} \sqrt{1 + |\nabla u(x, y)|^2} \, dx \, dy,$$

where $u : \Omega \rightarrow \mathbb{R}$ is a graph over a region $\Omega \subset \mathbb{R}^2$. Minimizing $\mathcal{A}$ among all admissible surfaces with the same boundary leads to the *minimal surface equation*, which states that the surface has zero mean curvature. This variational formulation underlies both theoretical results, such as existence and regularity of solutions (the Plateau problem), and modern numerical methods for approximating minimal surfaces. (For physical soap films formed on wire frames, this minimal area property is directly observed as the film attempts to reduce its surface energy in response to surface tension.)

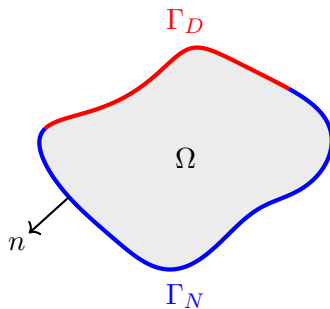
1.1 Functionals and variational problems

In classical calculus, the derivative of a real-valued function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto f(x),$$

is defined as

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$


 Figure 1.1: A domain Ω with its boundary $\Gamma = \Gamma_D \cup \Gamma_N$

If $f \in C^2(\mathbb{R})$ and $f'(x_0) = 0$, the point x_0 is called a *critical point*. It corresponds to a local extremum if, in a neighborhood $|x - x_0| < \delta$,

$$\begin{cases} f(x) \leq f(x_0), & \text{if } f''(x_0) < 0 \quad (\text{maximum}), \\ f(x) \geq f(x_0), & \text{if } f''(x_0) > 0 \quad (\text{minimum}). \end{cases}$$

The *calculus of variations* generalizes this framework to situations where the quantity to be optimized is no longer a function of real variables, but a mapping whose argument is itself a function. Such mappings are called *functionals*.

A functional is an application of the form

$$\mathcal{J} : U \longrightarrow \mathbb{R},$$

where U is a space of admissible functions. Typical variational problems consist in finding a function $u \in U$ such that

$$\mathcal{J}(u) = \min_{v \in U} \mathcal{J}(v),$$

or, more generally, such that $\mathcal{J}$ is stationary at u .

Variational principles arise naturally in physics and engineering, where fundamental laws are often expressed as extremality or stationarity conditions on energies, actions, or geometric quantities.

Example: the Poisson problem and Dirichlet energy Let $\Omega \subset \mathbb{R}^d$ ($d = 1, 2, 3$) be a bounded domain with boundary $\Gamma = \Gamma_D \cup \Gamma_N$ (see Figure 1.1). We consider a scalar field $u(\mathbf{x})$ subjected to prescribed values $u = \bar{u}$ (Dirichlet boundary conditions) on Γ_D and to fluxes $\kappa \nabla u \cdot \mathbf{n} = \bar{q}$ (Neumann boundary conditions) on Γ_N . The *strong form* of the Poisson problem reads

$$\begin{aligned} -\nabla \cdot (\kappa \nabla u) &= f & \text{in } \Omega, \\ u &= \bar{u} & \text{on } \Gamma_D \\ \kappa \nabla u \cdot \mathbf{n} &= \bar{q} & \text{on } \Gamma_N \end{aligned} \tag{1.1}$$

where $\kappa(\mathbf{x}) > 0$ is a given diffusion coefficient and $f(\mathbf{x})$ is a source term. We introduce the *space of admissible functions*

$$U = \{u \in H^1(\Omega) \mid u|_{\Gamma_D} = \bar{u}\}, \tag{1.2}$$

where $\bar{u}$ denotes the prescribed Dirichlet boundary data. The *Dirichlet energy* associated with the Poisson problem is defined by the *functional*

$$\Pi(u) = \frac{1}{2} \int_{\Omega} \kappa |\nabla u|^2 dv - \int_{\Omega} f u dv - \int_{\Gamma_N} \bar{q} u ds, \tag{1.3}$$

where $\bar{q}$ denotes the prescribed Neumann flux on Γ_N . The space $H^1(\Omega)$ will be defined rigorously in a forthcoming chapter. At this stage, it is sufficient to observe that $H^1(\Omega)$ is the natural energy space for the Poisson problem, namely the space of functions for which the Dirichlet energy $\Pi(u)$ is finite. The Poisson problem can then be formulated as the *variational problem*:

$$\text{Find } u \in U \text{ such that } \Pi(u) \leq \Pi(v) \quad \forall v \in U.$$

At this stage, it is not yet possible to establish a direct link between the minimization of the Dirichlet energy and the Poisson problem written in strong form. The purpose of the following developments, which constitute the core of the calculus of variations, is precisely to make this connection explicit. The first thing that must be noticed is that the argument of the functional Π is not a real number but a function, and classical differentiation is no longer applicable.

1.2 First variation and stationarity

Let $u \in U$. A variation δu is a function such that

$$u + \delta u \in U.$$

This implies that δu vanishes on Γ_D because every function of U has the same value $\bar{u}$ on Γ_D . The space of admissible variations is therefore

$$U_0 = \{ \delta u \in H^1(\Omega) \mid \delta u|_{\Gamma_D} = 0 \}. \quad (1.4)$$

Let $\mathcal{J} : U \rightarrow \mathbb{R}$ be a functional. The *first variation* of $\mathcal{J}$ at $u \in U$ in the direction $\delta u \in U_0$ is defined as the directional (Gâteaux) derivative

$$\delta \mathcal{J}(u; \delta u) := \lim_{\varepsilon \rightarrow 0} \frac{\mathcal{J}(u + \varepsilon \delta u) - \mathcal{J}(u)}{\varepsilon},$$

provided that this limit exists. The functional $\mathcal{J}$ is said to be *stationary* at $u \in U$ if

$$\delta \mathcal{J}(u; \delta u) = 0 \quad \forall \delta u \in U_0.$$

This condition is the natural infinite-dimensional analogue of the vanishing of the derivative in classical calculus.

1.2.1 From stationarity to the Euler–Lagrange equation

We now explain how the stationarity condition

$$\delta \mathcal{J}(u; \delta u) = 0 \quad \forall \delta u \in U_0$$

leads to a strong form i.e. a differential equation. For clarity, we first work in one space dimension and consider a functional of the form

$$\mathcal{J}(y) = \int_a^b F(x, y(x), y'(x)) dx, \quad (1.5)$$

where F is assumed to be sufficiently smooth with respect to its arguments. The admissible set is

$$U = \{ y \in C(a, b) \mid y(a) = \alpha, y(b) = \beta \}, \quad U_0 = \{ \delta y \in C(a, b) \mid \delta y(a) = 0, \delta y(b) = 0 \}$$

where C is a space of functions that is such $\mathcal{J}(y)$ makes sense.

Fundamental lemma of the calculus of variations. A key result underlying the derivation of the Euler–Lagrange equations is the so-called *fundamental lemma of the calculus of variations*.

Let f be a continuous function on the interval $[a, b]$. If

$$\int_a^b f(x) w(x) dx = 0 \quad \text{for all smooth functions } w \text{ such that } w(a) = w(b) = 0,$$

then

$$f(x) = 0 \quad \text{for all } x \in (a, b).$$

This result shows that a function whose integral against all admissible variations vanishing at the boundary is zero must itself vanish identically. It provides the rigorous justification for passing from a variational statement to a pointwise differential equation.

At this stage, we deliberately avoid introducing the theory of distributions and the space of smooth functions with compact support. These notions will be presented later, when they become necessary for a more general and rigorous functional framework.

In the following, we will use this lemma to deduce the Euler–Lagrange equations from the stationarity condition of a functional.

First variation. Let $y \in U$ and $\delta y \in U_0$. Define $\phi(\varepsilon) = \mathcal{J}(y + \varepsilon \delta y)$. By the chain rule,

$$\phi'(\varepsilon) = \int_a^b \left(\frac{\partial F}{\partial y}(x, y + \varepsilon \delta y, y' + \varepsilon \delta y') \delta y + \frac{\partial F}{\partial y'}(x, y + \varepsilon \delta y, y' + \varepsilon \delta y') \delta y' \right) dx.$$

Hence the first variation is

$$\delta \mathcal{J}(y; \delta y) = \phi'(0) = \int_a^b \left(\frac{\partial F}{\partial y}(x, y, y') \delta y + \frac{\partial F}{\partial y'}(x, y, y') \delta y' \right) dx. \quad (1.6)$$

Integration by parts. Using integration by parts and the fact that $\delta y(a) = \delta y(b) = 0$, we obtain

$$\begin{aligned} \delta \mathcal{J}(y; \delta y) &= \int_a^b \frac{\partial F}{\partial y}(x, y, y') \delta y dx + \left[\frac{\partial F}{\partial y'}(x, y, y') \delta y \right]_a^b - \int_a^b \frac{d}{dx} \left(\frac{\partial F}{\partial y'}(x, y, y') \right) \delta y dx \\ &= \int_a^b \left(\frac{\partial F}{\partial y}(x, y, y') - \frac{d}{dx} \left(\frac{\partial F}{\partial y'}(x, y, y') \right) \right) \delta y dx. \end{aligned} \quad (1.7)$$

Fundamental lemma and Euler–Lagrange equation. Assume y is stationary, i.e.

$$\delta \mathcal{J}(y; \delta y) = 0 \quad \forall \delta y \in U_0.$$

By the fundamental lemma of the calculus of variations, the integrand in (1.7) must vanish almost everywhere in (a, b) . Therefore, y satisfies the *Euler–Lagrange equation*

$$\boxed{\frac{\partial F}{\partial y}(x, y, y') - \frac{d}{dx} \left(\frac{\partial F}{\partial y'}(x, y, y') \right) = 0 \quad \text{in } (a, b).} \quad (1.8)$$

Remark (natural boundary conditions). If the endpoint values are not prescribed (so that δy does not necessarily vanish at a and/or b), then the boundary term in (1.7) yields additional *natural* boundary conditions. For instance, if $y(a)$ is free, stationarity implies

$$\left. \frac{\partial F}{\partial y'}(x, y, y') \right|_{x=a} = 0,$$

and similarly at $x = b$.

Beltrami identity. In the special case where the integrand F in (1.5) does not depend explicitly on the independent variable x , that is,

$$F = F(y, y'),$$

the Euler–Lagrange equation admits a first integral known as the *Beltrami identity*. Indeed, since

$$\frac{dF}{dx} = \frac{\partial F}{\partial y} y' + \frac{\partial F}{\partial y'} y'',$$

and using the Euler–Lagrange equation

$$\frac{\partial F}{\partial y} = \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right),$$

we obtain

$$\frac{dF}{dx} = \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) y' + \frac{\partial F}{\partial y'} y'' = \frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} \right).$$

Therefore,

$$\frac{d}{dx} \left(F - y' \frac{\partial F}{\partial y'} \right) = 0,$$

which implies that

$$\boxed{F - y' \frac{\partial F}{\partial y'} = C,} \quad (1.9)$$

where C is a constant.

The Beltrami identity provides a conserved quantity and often allows the Euler–Lagrange equation to be reduced from a second-order to a first-order differential equation. It plays a key role in several classical variational problems, including the brachistochrone and geodesic problems.

1.3 From Dirichlet energy to the Poisson equation

Step 1: first variation (Gâteaux derivative). Consider the Dirichlet energy functional (1.3) and the space U of admissible functions (see (1.2)) as well as the space U_0 of variations (see (1.4)). Let $u \in U$ and $\delta u \in U_0$. Define $\phi(\varepsilon) = \Pi(u + \varepsilon \delta u)$. Using $\nabla(u + \varepsilon \delta u) = \nabla u + \varepsilon \nabla \delta u$ and expanding,

$$\phi(\varepsilon) = \frac{1}{2} \int_{\Omega} \kappa |\nabla u + \varepsilon \nabla \delta u|^2 dx - \int_{\Omega} f(u + \varepsilon \delta u) dx - \int_{\Gamma_N} \bar{q}(u + \varepsilon \delta u) ds.$$

Since

$$|\nabla u + \varepsilon \nabla \delta u|^2 = |\nabla u|^2 + 2\varepsilon \nabla u \cdot \nabla \delta u + \varepsilon^2 |\nabla \delta u|^2,$$

we obtain

$$\phi(\varepsilon) = \Pi(u) + \varepsilon \left(\int_{\Omega} \kappa \nabla u \cdot \nabla \delta u dx - \int_{\Omega} f \delta u dx - \int_{\Gamma_N} \bar{q} \delta u ds \right) + \frac{\varepsilon^2}{2} \int_{\Omega} \kappa |\nabla \delta u|^2 dx.$$

Therefore,

$$\boxed{\delta \Pi(u; \delta u) = \phi'(0) = \int_{\Omega} \kappa \nabla u \cdot \nabla \delta u dx - \int_{\Omega} f \delta u dx - \int_{\Gamma_N} \bar{q} \delta u ds.} \quad (1.10)$$

In particular, the stationarity condition $\delta \Pi(u; \delta u) = 0 \forall \delta u \in U_0$ yields the classical *weak formulation* that is used in finite elements.

Step 2: integration by parts (Green's formula). Assume for the moment that u is smooth enough (e.g. $u \in C^2(\Omega)$) so that the computations below are classical. Using the identity

$$\nabla \cdot (\kappa \delta u \nabla u) = \kappa \nabla u \cdot \nabla \delta u + \delta u \nabla \cdot (\kappa \nabla u),$$

we get, by integrating over Ω and applying the divergence theorem,

$$\int_{\Omega} \kappa \nabla u \cdot \nabla \delta u \, dx = \int_{\Gamma} \kappa \nabla u \cdot n \delta u \, ds - \int_{\Omega} \delta u \nabla \cdot (\kappa \nabla u) \, dx.$$

Split $\Gamma = \Gamma_D \cup \Gamma_N$. Since $\delta u|_{\Gamma_D} = 0$, the boundary term reduces to

$$\int_{\Gamma} \kappa \nabla u \cdot n \delta u \, ds = \int_{\Gamma_N} \kappa \nabla u \cdot n \delta u \, ds.$$

Substituting into (1.10), the stationarity condition becomes

$$\int_{\Omega} \left(-\nabla \cdot (\kappa \nabla u) - f \right) \delta u \, dx + \int_{\Gamma_N} \left(\kappa \nabla u \cdot n - \bar{q} \right) \delta u \, ds = 0 \quad \forall \delta u \in U_0. \quad (1.11)$$

Step 3: conclude with the fundamental lemma. Choose test functions

$$\delta u \in U'_0 = \{ \delta u \in H^1(\Omega) \mid \delta u|_{\Gamma} = 0 \} \subset U_0,$$

then the boundary term in (1.11) vanishes and we obtain

$$\int_{\Omega} \left(-\nabla \cdot (\kappa \nabla u) - f \right) \delta u \, dx = 0 \quad \forall \delta u \in U'_0.$$

By the fundamental lemma of the calculus of variations,

$$-\nabla \cdot (\kappa \nabla u) = f \quad \text{in } \Omega.$$

Next, allowing general $\delta u \in U_0$ and using trace arguments, (1.11) implies the natural boundary condition on Γ_N ,

$$\kappa \nabla u \cdot n = \bar{q} \quad \text{on } \Gamma_N,$$

while the definition of U enforces the essential boundary condition

$$u = \bar{u} \quad \text{on } \Gamma_D.$$

The variational principle (1.10) is thus equivalent to the strong form of the Poisson problem (1.2). Note that, in order to recover the strong (differential) form of the problem from its variational formulation, we have implicitly assumed a rather strong regularity of the solution u . In particular, the strong form requires u to be at least twice differentiable, i.e. $u \in C^2(\Omega)$, so that all derivatives appearing in the differential equation are well defined pointwise.

By contrast, the weak formulation only requires $u \in H^1(\Omega)$. As will be shown in a forthcoming chapter, the Sobolev space $H^1(\Omega)$ is strictly larger than $C^2(\Omega)$: every classical solution is a weak solution, but the converse is not true. Weak solutions may exist even when the solution is not twice differentiable, or not even differentiable in the classical sense everywhere. This is not a mathematical artifact but rather a physically meaningful feature of the model.

A typical situation where this distinction is essential arises when the diffusion coefficient κ is discontinuous. For instance, consider two materials with different conductivities that are perfectly bonded together. In this case, κ exhibits jump discontinuities across material interfaces. The weak formulation naturally enforces the continuity of the flux $\kappa \partial_x u$, which expresses the

balance of physical fluxes across the interface. However, the derivative $\partial_x u$ itself generally becomes discontinuous at the interface. As a consequence, the solution u cannot belong to $C^2(\Omega)$, and the strong form of the equation is no longer meaningful in the classical sense.

The variational formulation remains well posed in this setting and admits solutions in $H^1(\Omega)$, whose limited regularity accurately reflects the underlying physics. This robustness with respect to irregular coefficients and reduced regularity is one of the main reasons why weak formulations, and the finite element methods derived from them, are preferred over strong formulations for both analysis and numerical simulation.

Conclusion (Euler–Lagrange equation for Π). The stationarity of Π over U yields the weak form

$$\int_{\Omega} \kappa \nabla u \cdot \nabla \delta u \, dx = \int_{\Omega} f \delta u \, dx + \int_{\Gamma_N} \bar{q} \delta u \, ds \quad \forall \delta u \in U_0,$$

and, under sufficient regularity, the corresponding strong (Euler–Lagrange) equations

$$\begin{aligned} -\nabla \cdot (\kappa \nabla u) &= f && \text{in } \Omega, \\ u &= \bar{u} && \text{on } \Gamma_D, \\ \kappa \nabla u \cdot n &= \bar{q} && \text{on } \Gamma_N. \end{aligned}$$

Remark (second variation and convexity). For any admissible variation $\delta u \in U_0$, the second variation of the Dirichlet energy reads

$$\delta^2 \Pi(u; \delta u, \delta u) = \int_{\Omega} \kappa |\nabla \delta u|^2 \, dx \geq 0.$$

If $\kappa > 0$ almost everywhere in Ω , this quadratic form is strictly positive for all nonzero δu , which implies that the functional Π is strictly convex on U . As a consequence, the stationary point of Π is not only a critical point but the unique global minimizer.

It is important to emphasize, however, that this conclusion relies crucially on the convexity of the functional. In more general variational problems, a vanishing first variation merely characterizes stationary points, which may correspond to minima, maxima, or saddle points. It is perfectly possible for a functional to possess a unique stationary point that does *not* correspond to a minimum.

The calculus of variations provides a systematic framework to derive the Euler–Lagrange equations governing stationary points of a functional. It does not, by itself, guarantee that these stationary points minimize (or maximize) the functional. Establishing the nature of a stationary point requires additional information, such as the analysis of the second variation, convexity properties, or more refined energetic arguments. In this sense, the calculus of variations is a fundamental tool for deriving governing equations, but it does not necessarily imply that the underlying problem is a minimization problem.

1.4 Classical Examples

1.4.1 The brachistochrone problem

The brachistochrone problem is a classical variational problem that historically motivated the development of the calculus of variations. Given two points $A = (0, y_0)$ and $B = (x_1, 0)$ with $x_1 > 0$, we seek a curve $y = y(x)$ in the vertical plane joining A to B such that a point mass, released from rest at A and sliding without friction under uniform gravity, reaches B in minimal time (see Figure 1.2). Throughout this section we assume that gravity acts *downwards*

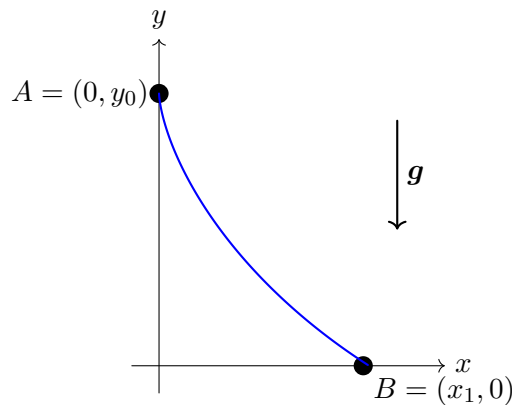


Figure 1.2: Brachistochrone problem.

Time of descent as a functional. Let the curve be parameterized as $y = y(x)$, $x \in [0, x_1]$, with

$$y(0) = y_0, \quad y(x_1) = 0.$$

An infinitesimal arc-length element is

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + (y')^2} dx, \quad y' = \frac{dy}{dx}.$$

If v denotes the speed of the particle, the travel time satisfies

$$dt = \frac{ds}{v}, \quad T = \int_0^{x_1} \frac{ds}{v}.$$

Since the particle starts from rest at $y = y_0$ and friction is neglected, conservation of mechanical energy gives

$$mg(y_0 - y) = \frac{1}{2}mv^2, \quad \Rightarrow \quad v = \sqrt{2g(y_0 - y)},$$

where $g > 0$ is the gravitational acceleration. Therefore, the time of descent can be written as the *functional*

$$T(y) = \int_0^{x_1} \sqrt{\frac{1 + (y')^2}{2g(y_0 - y)}} dx. \quad (1.12)$$

This is of the form $\mathcal{J}(y) = \int_a^b F(x, y, y') dx$ with

$$F(x, y, y') = \sqrt{\frac{1 + (y')^2}{2g(y_0 - y)}}.$$

Not that F does not explicitly depends on x so we can use Beltrami's identity (1.9) for solving the brachistochrone problem.

Admissible curves and variations We define the admissible set

$$U = \{y \in H^1(0, x_1) \mid y(0) = y_0, y(x_1) = 0, y(x) > 0 \text{ on } (0, x_1]\}.$$

In one dimension, every function $u \in H^1(0, x_1)$ admits a uniformly continuous representative on the closed interval $[0, x_1]$.

A variation δy is an admissible perturbation direction such that $y + \varepsilon \delta y \in U$ for $|\varepsilon|$ small. In particular, endpoint constraints imply

$$\delta y(0) = 0, \quad \delta y(x_1) = 0.$$

We thus introduce the space of variations

$$U_0 = \{ \delta y \in H^1(0, x_1) \mid \delta y(0) = 0, \delta y(x_1) = 0 \}.$$

First variation and Euler–Lagrange equation Let $y \in U$ and $\delta y \in U_0$. The first variation of T at y in the direction δy is the Gâteaux derivative

$$\delta T(y; \delta y) = \left. \frac{d}{d\varepsilon} T(y + \varepsilon \delta y) \right|_{\varepsilon=0}.$$

For a functional of the form $\int F(x, y, y') dx$, the standard computation yields

$$\delta T(y; \delta y) = \int_0^{x_2} \left(\frac{\partial F}{\partial y} \delta y + \frac{\partial F}{\partial y'} \delta y' \right) dx.$$

Integrating by parts the term involving $\delta y'$ and using the endpoint conditions $\delta y(0) = \delta y(x_2) = 0$ gives

$$\delta T(y; \delta y) = \int_0^{x_2} \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right) \delta y dx.$$

Hence a minimizer (or more generally a stationary curve) satisfies the Euler–Lagrange equation

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \quad \text{in } (0, x_2).} \quad (1.13)$$

Beltrami identity and first integral In the brachistochrone problem, F does not depend explicitly on x . Therefore, the Euler–Lagrange equation admits the first integral (Beltrami identity)

$$\boxed{F - y' \frac{\partial F}{\partial y'} = C,} \quad (1.14)$$

where C is a constant. For

$$F(y, y') = \sqrt{\frac{1 + (y')^2}{2g(y_0 - y)}}, \quad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{2g(y_0 - y)} \sqrt{1 + (y')^2}},$$

and thus

$$F - y' \frac{\partial F}{\partial y'} = \frac{1}{\sqrt{2g(y_0 - y)}} \left(\sqrt{1 + (y')^2} - \frac{(y')^2}{\sqrt{1 + (y')^2}} \right) = \frac{1}{\sqrt{2g(y_0 - y)}} \frac{1}{\sqrt{1 + (y')^2}}.$$

Beltrami's identity (1.14) therefore yields

$$\frac{1}{\sqrt{2g(y_0 - y)} \sqrt{1 + (y')^2}} = C,$$

or equivalently

$$\boxed{(1 + (y')^2) (y_0 - y) = \frac{1}{2gC^2} =: D, \quad D > 0.} \quad (1.15)$$

This non linear ordinary differential equation has an analytical solution that we will fully develop.

Parametric solution: the cycloid. Equation (1.15) can be integrated by introducing the substitution

$$y' = \tan \theta,$$

so that $1 + (y')^2 = \sec^2 \theta$. Equation (1.15) becomes

$$(y_0 - y) \sec^2 \theta = D \quad \Rightarrow \quad y = y_0 - D \cos^2 \theta = \frac{D}{2} (1 + \cos(2\theta)).$$

Set $u = 2\theta$ and define $R = D/2$. Then

$$\boxed{y = y_0 - R(1 + \cos u).} \quad (1.16)$$

To obtain x , note that

$$y' = \frac{dy}{dx} = \tan \theta \quad \Rightarrow \quad \frac{dx}{d\theta} = \frac{dx}{dy} \frac{dy}{d\theta} = \frac{dy}{d\theta} \frac{1}{\tan \theta}.$$

Since $y = D \cos^2 \theta$,

$$\frac{dy}{d\theta} = 2D \sin \theta \cos \theta,$$

hence

$$\frac{dx}{d\theta} = \frac{2D \sin \theta \cos \theta}{\sin \theta / \cos \theta} = 2D \cos^2 \theta.$$

With $u = 2\theta$ and $R = D/2$, we get

$$\frac{dx}{du} = \frac{dx}{d\theta} \frac{d\theta}{du} = (2D \cos^2 \theta) \frac{1}{2} = D \cos^2 \theta = 2R \cos^2 \theta = R(1 + \cos u).$$

Integrating in u gives

$$\boxed{x = R(u + \sin u) + c.} \quad (1.17)$$

Equations (1.16)–(1.17) are the parametric equations of a cycloid (up to a choice of orientation). The cycloid is the trajectory of a material point attached to the rim of a circular wheel rolling without slipping on a horizontal surface, combining translational and rotational motion.

Determination of the constants. Recall the parametric equations of the cycloid:

$$\begin{aligned} x(u) &= R(u + \sin u) + c, \\ y(u) &= y_0 - R(1 + \cos u). \end{aligned}$$

We must determine $R > 0$ and c so that the curve passes through $A = (0, y_0)$ and $B = (x_1, 0)$. If $y = y_0$, $1 + \cos u = 0$, i.e. $u = \pi \pmod{2\pi}$. If $y = y_0$, $x = 0$ so choosing $u = \pi$ in $x(u)$ yields

$$0 = R(\pi + \sin \pi) + c \quad \Rightarrow \quad c = R\pi.$$

Then, the parameter value $u = u_1$ corresponding to point B satisfies

$$x_1 = R(u_1 + \sin u_1), \quad y_0 = R(1 + \cos u_1),$$

which is a system of two non linear equations with two unknowns R and u_1 that can be solved numerically. For example, if $y_0 = x_1 = 1$, we have to solve

$$1 = R(1 - \cos(u_1)) \quad \rightarrow \quad R = \frac{1}{1 - \cos u_1}.$$

Then,

$$1 - \cos u_1 = u_1 - \sin(u_1)$$

that is an equation that must be resolved numerically and the solution is $u_1 \approx 2.412$. Then, $R \approx 0.573$. Figure 1.3 shows three brachistochrone curves.

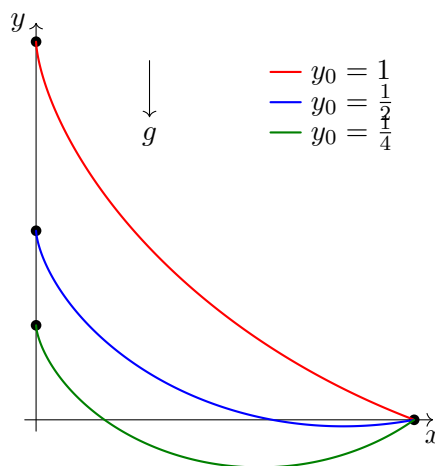


Figure 1.3: Brachistochrone curves for gravity acting downward, connecting $(0, y_0)$ to $(x_1, 0)$ for $x_1 = 1$ and $y_0 \in \{1, \frac{1}{2}, \frac{1}{4}\}$.

Tautochrone property of a cycloid. We consider again the cycloid

$$x(u) = R(u - \sin u), \quad y(u) = y_0 - R(1 - \cos u), \quad u \in [0, u_1],$$

with $R > 0$. On one arch $u \in [0, \pi]$, the lowest point is at $u = \pi$, since

$$y'(u) = -R \sin u, \quad y'(\pi) = 0, \quad y''(\pi) = -R \cos \pi = R > 0.$$

Release the bead from rest at some point $u = u_0 \in (0, \pi)$, whose height is

$$y_1 := y(u_0) = y_0 - R(1 - \cos u_0) < y_0.$$

Energy conservation between the release point and a generic point $u \in [u_0, \pi]$ gives

$$\frac{1}{2}mv(u)^2 + mg y(u) = mg y(u_0) \quad \implies \quad v(u) = \sqrt{2g(y(u_0) - y(u))}. \quad (1.18)$$

Using the parametrization,

$$y(u_0) - y(u) = (y_0 - R(1 - \cos u_0)) - (y_0 - R(1 - \cos u)) = R(\cos u_0 - \cos u),$$

hence

$$v(u) = \sqrt{2Rg(\cos u_0 - \cos u)}. \quad (1.19)$$

In half-angle form,

$$\cos u_0 - \cos u = 2\left(\sin^2 \frac{u}{2} - \sin^2 \frac{u_0}{2}\right),$$

so

$$v(u) = 2\sqrt{Rg} \sqrt{\sin^2\left(\frac{u}{2}\right) - \sin^2\left(\frac{u_0}{2}\right)}. \quad (1.20)$$

Compute arclength as

$$x'(u) = R(1 - \cos u), \quad y'(u) = -R \sin u,$$

so

$$ds = \sqrt{x'(u)^2 + y'(u)^2} du = R\sqrt{(1 - \cos u)^2 + \sin^2 u} du = R\sqrt{2 - 2\cos u} du \quad (1.21)$$

$$= 2R \sin\left(\frac{u}{2}\right) du, \quad (\text{since } u \in [0, \pi] \Rightarrow \sin(u/2) \geq 0). \quad (1.22)$$

We show now that the time of descent to the bottom is independent of u_0 . The time element is $dt = ds/v$. Using (1.20),

$$dt = \frac{2R \sin(u/2) du}{2\sqrt{Rg} \sqrt{\sin^2(u/2) - \sin^2(u_0/2)}} = \sqrt{\frac{R}{g}} \frac{\sin(u/2) du}{\sqrt{\sin^2(u/2) - \sin^2(u_0/2)}}. \quad (1.23)$$

Therefore the time to reach the lowest point $u = \pi$ from any release point $u_0 \in (0, \pi)$ is

$$T(u_0 \rightarrow \pi) = \sqrt{\frac{R}{g}} \int_{u_0}^{\pi} \frac{\sin(u/2)}{\sqrt{\sin^2(u/2) - \sin^2(u_0/2)}} du. \quad (1.24)$$

Set

$$w = \sin\left(\frac{u}{2}\right), \quad w_0 = \sin\left(\frac{u_0}{2}\right),$$

so $du = \frac{2 dw}{\sqrt{1-w^2}}$ and (1.24) becomes

$$T(u_0 \rightarrow \pi) = \sqrt{\frac{R}{g}} 2 \int_{w_0}^1 \frac{w}{\sqrt{w^2 - w_0^2} \sqrt{1-w^2}} dw. \quad (1.25)$$

Now use the substitution

$$w^2 = w_0^2 + (1 - w_0^2) \sin^2 t, \quad t \in [0, \pi/2].$$

Then

$$dw = \frac{(1 - w_0^2) \sin t \cos t}{w} dt, \quad \sqrt{w^2 - w_0^2} = \sqrt{1 - w_0^2} \sin t, \quad \sqrt{1 - w^2} = \sqrt{1 - w_0^2} \cos t,$$

and the integrand simplifies to dt . The limits are $w = w_0 \Rightarrow t = 0$ and $w = 1 \Rightarrow t = \pi/2$. Hence

$$2 \int_{w_0}^1 \frac{w}{\sqrt{w^2 - w_0^2} \sqrt{1 - w^2}} dw = 2 \int_0^{\pi/2} dt = \pi.$$

Therefore

$$\boxed{T(u_0 \rightarrow \pi) = \pi \sqrt{\frac{R}{g}}}, \quad (1.26)$$

which is *independent* of the release point u_0 (equivalently, independent of the release height $y_1 < y_0$).

Remarkably, the motion of a particle along a cycloid is dynamically equivalent to that of a simple harmonic oscillator, with a natural angular frequency proportional to $\sqrt{g/R}$, which coincides with the angular frequency of a simple pendulum of length R in a uniform gravitational field.

Conclusion. For the cycloid above, every bead released from rest at any point on the arch reaches the lowest point in the same time. This is precisely the tautochrone (isochronous) property.

1.4.2 The isoperimetric problem

The isoperimetric problem is one of the oldest problems in the calculus of variations. It consists in finding, among all closed plane curves of given length, the one that encloses the maximum area.

Geometric setting Let a simple, closed, sufficiently smooth plane curve be parametrized by

$$\mathbf{x}(s) = (x(s), y(s)), \quad s \in [0, 1],$$

with periodicity conditions

$$\mathbf{x}(0) = \mathbf{x}(1), \quad \mathbf{x}'(0) = \mathbf{x}'(1).$$

The length of the curve is

$$L(\mathbf{x}) = \int_0^1 \sqrt{x'(s)^2 + y'(s)^2} ds, \quad (1.27)$$

and the area enclosed by the curve is given by Green's formula:

$$A(\mathbf{x}) = \frac{1}{2} \int_0^1 (x y' - y x') ds. \quad (1.28)$$

Variational formulation with constraint We seek to maximize the area $A(\mathbf{x})$ under the constraint that the length $L(\mathbf{x})$ is prescribed:

$$L(\mathbf{x}) = L_0.$$

This constrained problem is handled by introducing a Lagrange multiplier λ and defining the augmented functional

$$\mathcal{J}(\mathbf{x}) = A(\mathbf{x}) - \lambda L(\mathbf{x}). \quad (1.29)$$

A stationary curve satisfies

$$\delta \mathcal{J}(\mathbf{x}; \delta \mathbf{x}) = 0 \quad \forall \delta \mathbf{x}.$$

First variation Using (1.28) and (1.27), the integrand of $\mathcal{J}$ can be written as

$$F(x, y, x', y') = \frac{1}{2}(xy' - yx') - \lambda \sqrt{x'^2 + y'^2}.$$

Applying the Euler–Lagrange equations to x and y yields the system

$$-\frac{1}{2}y' - \lambda \frac{d}{ds} \left(\frac{x'}{\sqrt{x'^2 + y'^2}} \right) = 0, \quad (1.30)$$

$$\frac{1}{2}x' - \lambda \frac{d}{ds} \left(\frac{y'}{\sqrt{x'^2 + y'^2}} \right) = 0. \quad (1.31)$$

Differentiating the first equation with respect to s and combining both equations leads to

$$\frac{d}{ds} \left(\frac{x'}{\sqrt{x'^2 + y'^2}} \right) = -\kappa y', \quad \frac{d}{ds} \left(\frac{y'}{\sqrt{x'^2 + y'^2}} \right) = \kappa x', \quad (1.32)$$

where $\kappa = 1/(2\lambda)$ is constant.

Conclusion: the circle Equations (1.32) express that the curvature of the curve is constant. The only closed plane curves with constant curvature are circles. Therefore, the solution of the isoperimetric problem is a circle. In particular, among all plane curves of given length, the circle encloses the maximum possible area.

1.4.3 Fermat's principle and Snell's law

Another classical and fully analytical example of the calculus of variations is provided by geometrical optics. According to *Fermat's principle*, light travels between two points along the path that minimizes the travel time.

Geometric setting. Consider two homogeneous media separated by a flat interface $y = 0$. The refractive indices of the two media are $n_1 > 0$ for $y > 0$ and $n_2 > 0$ for $y < 0$. Let

$$A = (x_A, y_A), \quad y_A > 0,$$

$$B = (x_B, y_B), \quad y_B < 0,$$

be two fixed points. We assume that the light ray travels from A to B along a broken line that intersects the interface at a point

$$P = (x, 0),$$

which is free to vary.

Time of travel as a functional. In a homogeneous medium of refractive index n , the speed of light is c/n , where c is the speed of light in vacuum. The travel time is therefore proportional to the *optical length*. The total travel time from A to B through P is

$$T(x) = \frac{n_1}{c} \sqrt{(x - x_A)^2 + y_A^2} + \frac{n_2}{c} \sqrt{(x_B - x)^2 + y_B^2}. \quad (1.33)$$

Since the factor $1/c$ is constant, minimizing the travel time is equivalent to minimizing the functional

$$\mathcal{J}(x) = n_1 \sqrt{(x - x_A)^2 + y_A^2} + n_2 \sqrt{(x_B - x)^2 + y_B^2}.$$

Stationarity condition The minimizer x^* satisfies

$$\frac{d\mathcal{J}}{dx}(x^*) = 0.$$

Differentiating explicitly yields

$$n_1 \frac{x - x_A}{\sqrt{(x - x_A)^2 + y_A^2}} - n_2 \frac{x_B - x}{\sqrt{(x_B - x)^2 + y_B^2}} = 0.$$

Introduce the angles θ_1 and θ_2 formed by the ray with the normal to the interface in the first and second medium, respectively. By elementary geometry,

$$\sin \theta_1 = \frac{x - x_A}{\sqrt{(x - x_A)^2 + y_A^2}}, \quad \sin \theta_2 = \frac{x_B - x}{\sqrt{(x_B - x)^2 + y_B^2}}.$$

The stationarity condition therefore becomes

$$\boxed{n_1 \sin \theta_1 = n_2 \sin \theta_2.} \quad (1.34)$$

Conclusion Equation (1.34) is precisely the *Snell–Descartes law of refraction*. It shows that the laws of geometrical optics can be interpreted as Euler–Lagrange equations associated with a variational principle.

Remark. This example illustrates that the calculus of variations applies not only to mechanics and geometry, but also to wave propagation and optics. In more general settings, Fermat’s principle leads to curved light rays in inhomogeneous media, governed by nonlinear Euler–Lagrange equations.

1.4.4 The minimal surface problem

As in the brachistochrone problem, the minimal surface problem can be formulated as the minimization of a functional over a suitable class of admissible functions. We present here the classical setting of a surface given as the graph of a function $u(x, y)$.

Surface area as a functional Let $\omega \subset \mathbb{R}^2$ be a bounded domain with boundary $\partial\omega$. Consider surfaces in $\mathbb{R}^3$ of the form

$$S = \{(x, y, u(x, y)) \mid (x, y) \in \omega\},$$

where $u : \omega \rightarrow \mathbb{R}$ is sufficiently regular.

The area element of the graph is

$$dA = \sqrt{1 + |\nabla u|^2} \, dx \, dy, \quad \nabla u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right),$$

so the area of the surface is given by the functional

$$\boxed{\mathcal{A}(u) = \int_{\omega} \sqrt{1 + |\nabla u|^2} \, dx \, dy.} \quad (1.35)$$

Admissible surfaces and variations We prescribe the boundary of the surface by imposing a Dirichlet condition on $\partial\omega$: let $g : \partial\omega \rightarrow \mathbb{R}$ be given, and define the admissible set

$$U = \{u \mid u = g \text{ on } \partial\omega\}.$$

A variation δu must preserve the boundary data, hence it must vanish on $\partial\omega$:

$$U_0 = \{\delta u \mid \delta u = 0 \text{ on } \partial\omega\}.$$

For $u \in U$ and $\delta u \in U_0$, we consider perturbed surfaces $u_{\varepsilon} = u + \varepsilon \delta u$ and define

$$\delta \mathcal{A}(u; \delta u) = \left. \frac{d}{d\varepsilon} \mathcal{A}(u + \varepsilon \delta u) \right|_{\varepsilon=0}.$$

First variation Let

$$\phi(\varepsilon) = \mathcal{A}(u + \varepsilon \delta u) = \int_{\omega} \sqrt{1 + |\nabla u + \varepsilon \nabla \delta u|^2} \, dx \, dy.$$

Using the chain rule,

$$\phi'(\varepsilon) = \int_{\omega} \frac{(\nabla u + \varepsilon \nabla \delta u) \cdot \nabla \delta u}{\sqrt{1 + |\nabla u + \varepsilon \nabla \delta u|^2}} \, dx \, dy,$$

and therefore

$$\boxed{\delta \mathcal{A}(u; \delta u) = \int_{\omega} \frac{\nabla u \cdot \nabla \delta u}{\sqrt{1 + |\nabla u|^2}} \, dx \, dy.} \quad (1.36)$$

A minimizer (or more generally a stationary point) satisfies

$$\delta \mathcal{A}(u; \delta u) = 0 \quad \forall \delta u \in U_0.$$

Integration by parts and Euler–Lagrange equation Introduce the vector field

$$\mathbf{p}(u) = \frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}.$$

Then (1.36) becomes

$$\delta \mathcal{A}(u; \delta u) = \int_{\omega} \mathbf{p}(u) \cdot \nabla \delta u \, dx \, dy.$$

Assuming sufficient regularity, an integration by parts gives

$$\int_{\omega} \mathbf{p}(u) \cdot \nabla \delta u \, dx \, dy = - \int_{\omega} \nabla \cdot \mathbf{p}(u) \delta u \, dx \, dy + \int_{\partial\omega} (\mathbf{p}(u) \cdot \mathbf{n}) \delta u \, ds.$$

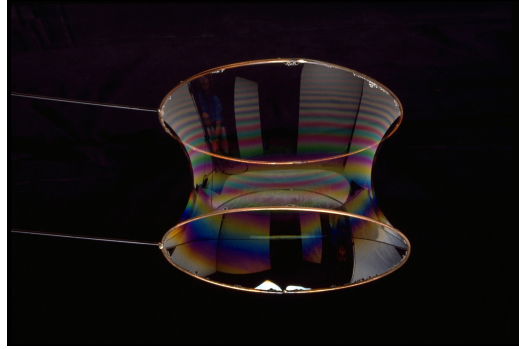


Figure 1.4: The catenoid obtained by revolving a catenary, observed experimentally as a soap film spanning two parallel circular rings.

Since $\delta u = 0$ on $\partial\omega$, the boundary term vanishes and the stationarity condition yields

$$\int_{\omega} \nabla \cdot \mathbf{p}(u) \delta u \, dx \, dy = 0 \quad \forall \delta u \in U_0.$$

Using the fundamental lemma (in the classical setting), we obtain the Euler–Lagrange equation of the minimal surface problem:

$$\boxed{\nabla \cdot \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0 \quad \text{in } \omega,} \quad (1.37)$$

together with the boundary condition

$$u = g \quad \text{on } \partial\omega.$$

Expanded form Writing (1.37) in coordinates yields the (non-linear) minimal surface equation

$$\boxed{(1 + u_y^2) u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2) u_{yy} = 0 \quad \text{in } \omega,} \quad (1.38)$$

where $u_x = \partial u / \partial x$, $u_y = \partial u / \partial y$, etc.

Remark. Equation (1.37) characterizes surfaces of zero mean curvature. In particular, planes are trivial solutions, and the equation is elliptic whenever $|\nabla u|$ is bounded.

1.4.5 Minimal surface of revolution: the catenoid between two circles

We now specialize the minimal surface problem to *surfaces of revolution*. This leads to a one-dimensional variational problem and yields the classical *catenoid* (see Figure 1.4).

Geometry and area functional Consider a surface obtained by revolving a planar curve $r = r(z) \geq 0$ around the z -axis. A convenient parametrization is

$$\mathbf{X}(z, \theta) = (r(z) \cos \theta, r(z) \sin \theta, z), \quad (z, \theta) \in [z_1, z_2] \times [0, 2\pi].$$

The area element is

$$|\partial_z \mathbf{X} \times \partial_\theta \mathbf{X}| = r(z) \sqrt{1 + r'(z)^2},$$

so the surface area is the functional

$$\boxed{\mathcal{A}(r) = 2\pi \int_{z_1}^{z_2} r(z) \sqrt{1 + r'(z)^2} \, dz.} \quad (1.39)$$

Admissible curves and variations (two circles) We prescribe that the surface spans two circles (coaxial with the z -axis). For instance, given radii $R_1, R_2 > 0$ at heights $z = z_1$ and $z = z_2$, we impose

$$r(z_1) = R_1, \quad r(z_2) = R_2.$$

Thus we consider admissible functions r satisfying these endpoint conditions, and variations δr such that

$$\delta r(z_1) = \delta r(z_2) = 0.$$

First variation and Euler–Lagrange equation The integrand in (1.39) is

$$F(r, r') = r\sqrt{1 + r'^2}.$$

As in the brachistochrone problem, stationarity of $\mathcal{A}$ yields the Euler–Lagrange equation

$$\boxed{\frac{\partial F}{\partial r} - \frac{d}{dz} \left(\frac{\partial F}{\partial r'} \right) = 0 \quad \text{in } (z_1, z_2).} \quad (1.40)$$

Here

$$\frac{\partial F}{\partial r} = \sqrt{1 + r'^2}, \quad \frac{\partial F}{\partial r'} = \frac{r r'}{\sqrt{1 + r'^2}},$$

hence (1.40) can be written as

$$\sqrt{1 + r'^2} - \frac{d}{dz} \left(\frac{r r'}{\sqrt{1 + r'^2}} \right) = 0.$$

Beltrami identity and reduction to first order Since $F(r, r')$ does not depend explicitly on z , we can use the Beltrami identity:

$$F - r' \frac{\partial F}{\partial r'} = C,$$

for some constant C . Substituting $F = r\sqrt{1 + r'^2}$ gives

$$r\sqrt{1 + r'^2} - r' \left(\frac{r r'}{\sqrt{1 + r'^2}} \right) = C \quad \implies \quad \boxed{\frac{r}{\sqrt{1 + r'^2}} = C.}$$

Equivalently,

$$1 + r'^2 = \frac{r^2}{C^2} \quad \implies \quad r'^2 = \frac{r^2}{C^2} - 1.$$

Integration: the catenoid Assuming $r > C > 0$ on (z_1, z_2) , we write

$$\frac{dr}{dz} = \sqrt{\frac{r^2}{C^2} - 1} \quad \implies \quad \frac{dr}{\sqrt{r^2 - C^2}} = \frac{dz}{C}.$$

Integrating yields

$$\operatorname{arcosh}\left(\frac{r}{C}\right) = \frac{z - z_0}{C},$$

for some constant z_0 , hence

$$\boxed{r(z) = C \cosh\left(\frac{z - z_0}{C}\right).} \quad (1.41)$$

Revolving this curve around the z -axis produces a *catenoid*, which is a minimal surface.

Fitting the two circles The constants $C > 0$ and z_0 are determined from the boundary conditions $r(z_1) = R_1$, $r(z_2) = R_2$, i.e.

$$R_1 = C \cosh\left(\frac{z_1 - z_0}{C}\right), \quad R_2 = C \cosh\left(\frac{z_2 - z_0}{C}\right).$$

A common symmetric case is two equal circles of radius R located at $z = \pm h$. By symmetry $z_0 = 0$, so the condition reduces to the single scalar equation

$$\boxed{R = C \cosh\left(\frac{h}{C}\right)}, \quad (1.42)$$

which determines C (when a solution exists). Introducing the dimensionless variable

$$s = \frac{h}{C}, \quad C = \frac{h}{s},$$

equation (1.42) can be rewritten as

$$\frac{R}{h} = \frac{\cosh s}{s}. \quad (1.43)$$

The function

$$\phi(s) = \frac{\cosh s}{s}, \quad s > 0,$$

admits a unique minimum. Differentiating,

$$\phi'(s) = \frac{s \sinh s - \cosh s}{s^2},$$

and the critical point satisfies

$$s \sinh s = \cosh s \quad \Longleftrightarrow \quad \tanh s = \frac{1}{s}. \quad (1.44)$$

Equation (1.44) has a unique solution

$$s_\star \approx 1.1996786403.$$

The minimal value of ϕ is therefore

$$\phi(s_\star) = \frac{\cosh s_\star}{s_\star} \approx 1.5088795615.$$

A catenoid spanning the two circles exists if and only if

$$\boxed{\frac{R}{h} \geq \phi(s_\star) \quad \Longleftrightarrow \quad \frac{h}{R} \leq 0.6627434193.} \quad (1.45)$$

Equivalently, in terms of the distance between the two circles $L = 2h$,

$$\boxed{\frac{L}{R} \leq 1.3254868386.}$$

At the critical configuration (1.45), the catenoid reaches a limiting shape with a vanishing second variation of the area functional. Beyond this threshold, no smooth minimal surface of revolution can connect the two circles. In the context of the Plateau problem, the area-minimizing configuration becomes the *Goldschmidt solution*, consisting of two disjoint planar disks.

Remark (degenerate limit). If one formally enforces the connectivity of the surface beyond the critical threshold, the catenoid degenerates into a non-regular configuration that may be interpreted as the limit of two cones meeting at a point. This configuration is not a solution of the Euler–Lagrange equation but represents the geometric limit of the family of catenoids.

1.4.6 A classical variational problem leading to a PDE: eigenmodes of a membrane

Let $\Omega \subset \mathbb{R}^2$ be a bounded domain with boundary Γ . We consider a thin membrane rigidly clamped along Γ . The vibration modes of the membrane are governed by an eigenvalue problem for the Laplacian (see LEPL1103). This eigenvalue problem can be obtained from a purely variational principle.

Rayleigh quotient and constrained minimization For a sufficiently smooth function u such that $u = 0$ on Γ , define the Dirichlet energy

$$E(u) = \int_{\Omega} |\nabla u|^2 dx,$$

and impose the normalization constraint

$$\int_{\Omega} u^2 dx = 1.$$

The *first eigenmode* is obtained by minimizing the energy under this constraint:

Find u such that $u|_{\Gamma} = 0$, $\int_{\Omega} u^2 dx = 1$, and $E(u) = \min$.

(1.46)

Equivalently, one may minimize the Rayleigh quotient

$$\mathcal{R}(u) = \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}, \quad u \neq 0, \quad u|_{\Gamma} = 0.$$

Variations and Lagrange multiplier Let u be a minimizer of (1.46). Admissible variations δu satisfy $\delta u = 0$ on Γ (to preserve the clamped boundary). To handle the constraint $\int_{\Omega} u^2 dx = 1$, we introduce a Lagrange multiplier λ and define the augmented functional

$$\mathcal{J}(u) = \int_{\Omega} |\nabla u|^2 dx - \lambda \left(\int_{\Omega} u^2 dx - 1 \right).$$

At a stationary point,

$$\delta \mathcal{J}(u; \delta u) = 0 \quad \forall \delta u \text{ with } \delta u|_{\Gamma} = 0.$$

First variation Compute the first variation term by term. Using the product rule,

$$\delta \left(\int_{\Omega} |\nabla u|^2 dx \right) = \int_{\Omega} 2 \nabla u \cdot \nabla(\delta u) dx, \quad \delta \left(\int_{\Omega} u^2 dx \right) = \int_{\Omega} 2u \delta u dx.$$

Therefore,

$$\delta \mathcal{J}(u; \delta u) = 2 \int_{\Omega} \nabla u \cdot \nabla(\delta u) dx - 2\lambda \int_{\Omega} u \delta u dx. \quad (1.47)$$

Dividing by 2 yields

$$\int_{\Omega} \nabla u \cdot \nabla(\delta u) dx - \lambda \int_{\Omega} u \delta u dx = 0 \quad \forall \delta u, \delta u|_{\Gamma} = 0.$$

Integration by parts and Euler–Lagrange PDE Assume u is smooth enough for the following calculation. By Green’s formula,

$$\int_{\Omega} \nabla u \cdot \nabla(\delta u) \, dx = - \int_{\Omega} (\Delta u) \delta u \, dx + \int_{\Gamma} \frac{\partial u}{\partial n} \delta u \, ds.$$

Since $\delta u = 0$ on Γ , the boundary term vanishes, and we obtain

$$\int_{\Omega} (-\Delta u - \lambda u) \delta u \, dx = 0 \quad \forall \delta u \text{ with } \delta u|_{\Gamma} = 0.$$

Using the fundamental lemma (in the classical setting), we conclude that

$$\boxed{-\Delta u = \lambda u \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \Gamma.} \quad (1.48)$$

Interpretation The parameter λ is an eigenvalue of the Laplacian with Dirichlet boundary conditions, and u is the associated eigenfunction. The smallest eigenvalue λ_1 corresponds to the minimizer of the Rayleigh quotient $\mathcal{R}(u)$, and higher modes can be obtained by adding orthogonality constraints with respect to previously found eigenfunctions.

1.4.7 A variational PDE: the Allen–Cahn equation (phase transitions)

We consider a scalar order parameter $u : \Omega \rightarrow \mathbb{R}$ on a bounded domain $\Omega \subset \mathbb{R}^d$. The Allen–Cahn model describes the competition between interfacial energy (penalizing gradients) and a double-well potential favoring the two phases $u \approx \pm 1$.

Energy functional Let $\varepsilon > 0$ be a small parameter and consider the Ginzburg–Landau (Allen–Cahn) energy

$$\boxed{\mathcal{E}(u) = \int_{\Omega} \left(\frac{\varepsilon}{2} |\nabla u|^2 + \frac{1}{\varepsilon} W(u) \right) dx, \quad W(u) = \frac{1}{4}(u^2 - 1)^2.} \quad (1.49)$$

We prescribe, for instance, homogeneous Dirichlet boundary conditions

$$u = 0 \quad \text{on } \Gamma,$$

or homogeneous Neumann boundary conditions

$$\frac{\partial u}{\partial n} = 0 \quad \text{on } \Gamma,$$

depending on the physical setting.

Admissible functions and variations Let U be the set of admissible functions satisfying the chosen boundary conditions. Variations δu must satisfy the corresponding homogeneous boundary conditions (e.g. $\delta u = 0$ on Γ in the Dirichlet case).

A stationary point $u \in U$ satisfies

$$\delta \mathcal{E}(u; \delta u) = 0 \quad \forall \delta u \in U_0.$$

First variation Let $u_{\eta} = u + \eta \delta u$. Using

$$\left. \frac{d}{d\eta} \right|_{\eta=0} \frac{\varepsilon}{2} |\nabla u_{\eta}|^2 = \varepsilon \nabla u \cdot \nabla(\delta u), \quad \left. \frac{d}{d\eta} \right|_{\eta=0} \frac{1}{\varepsilon} W(u_{\eta}) = \frac{1}{\varepsilon} W'(u) \delta u,$$

we obtain

$$\delta \mathcal{E}(u; \delta u) = \int_{\Omega} \left(\varepsilon \nabla u \cdot \nabla(\delta u) + \frac{1}{\varepsilon} W'(u) \delta u \right) dx.$$

Time-Dependent Allen–Cahn Equation in 2D

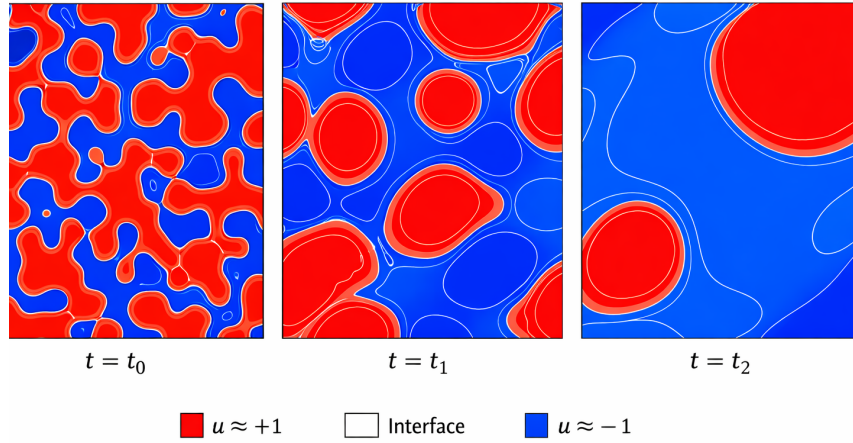


Figure 1.5: Temporal evolution of a phase transition using Allen-Kahn model.

Integration by parts and Euler–Lagrange equation Assuming sufficient regularity, integration by parts yields

$$\int_{\Omega} \varepsilon \nabla u \cdot \nabla(\delta u) dx = - \int_{\Omega} \varepsilon \Delta u \delta u dx + \int_{\Gamma} \varepsilon \frac{\partial u}{\partial n} \delta u ds.$$

The boundary term vanishes for admissible variations (either because $\delta u = 0$ on Γ in the Dirichlet case, or because $\partial u / \partial n = 0$ in the Neumann case). Hence stationarity implies

$$\int_{\Omega} \left(-\varepsilon \Delta u + \frac{1}{\varepsilon} W'(u) \right) \delta u dx = 0 \quad \forall \delta u,$$

and by the fundamental lemma we obtain the Euler–Lagrange PDE

$$\boxed{-\varepsilon \Delta u + \frac{1}{\varepsilon} W'(u) = 0 \quad \text{in } \Omega,} \quad (1.50)$$

with the corresponding boundary conditions. Since

$$W'(u) = u(u^2 - 1),$$

equation (1.50) reads

$$-\varepsilon \Delta u + \frac{1}{\varepsilon} u(u^2 - 1) = 0.$$

Remark (gradient flow). If instead of stationary points one considers the steepest descent of $\mathcal{E}$ in $L^2(\Omega)$, one obtains the time-dependent Allen–Cahn equation

$$\partial_t u = \varepsilon \Delta u - \frac{1}{\varepsilon} W'(u),$$

which is widely used to model curvature-driven phase separation (see Figure 1.5).

1.5 Ritz Method

The Ritz method is a variational technique for approximating the solution of boundary-value problems by restricting the search space to a finite-dimensional subspace of admissible functions. Starting from an energy (or variational) formulation of the problem, the exact solution is characterized as the minimizer of a functional over an infinite-dimensional space. The Ritz method consists in approximating this space by a finite-dimensional subspace spanned by chosen basis functions that satisfy the essential (Dirichlet) boundary conditions. The approximate solution is then obtained by minimizing the functional over this subspace, which leads to a finite system of algebraic equations. This method provides the theoretical foundation of the finite element method.

1.5.1 Example 1: Poisson problem in 1D.

Consider the Poisson equation on the interval $\Omega = (0, L)$:

$$-\frac{d^2 u}{dx^2} = 1 \quad \text{in } (0, L), \quad u(0) = u(L) = 0.$$

The associated energy functional is

$$\Pi(u) = \frac{1}{2} \int_0^L \left(\frac{du}{dx} \right)^2 dx - \int_0^L u dx,$$

defined on the space $H_0^1(0, L)$. The exact solution minimizes Π over this space.

In the Ritz method, we approximate u by

$$u_h(x) = \sum_{i=1}^N a_i \phi_i(x),$$

where the basis functions $\phi_i \in H_0^1(0, L)$ satisfy $\phi_i(0) = \phi_i(L) = 0$. Minimizing $\Pi(u_h)$ with respect to the coefficients a_i yields the linear system

$$\sum_{j=1}^N a_j \int_0^L \phi_i'(x) \phi_j'(x) dx = \int_0^L \phi_i(x) dx, \quad i = 1, \dots, N.$$

1.5.2 Example 2: Poisson problem in 2D.

Let $\Omega \subset \mathbb{R}^2$ be a square domain and consider

$$-\nabla^2 u = 1 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega.$$

The corresponding energy functional reads

$$\Pi(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 d\Omega - \int_{\Omega} u d\Omega,$$

defined on $H_0^1(\Omega)$. The solution of the Poisson problem is the unique minimizer of this functional.

Using the Ritz method, we seek an approximation of the form

$$u_h(x, y) = \sum_{i=1}^N a_i \phi_i(x, y),$$

where the basis functions $\phi_i \in H_0^1(\Omega)$ satisfy the homogeneous Dirichlet boundary conditions. The stationarity of $\Pi(u_h)$ with respect to a_i leads to the discrete system

$$\sum_{j=1}^N a_j \int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j d\Omega = \int_{\Omega} \phi_i d\Omega, \quad i = 1, \dots, N,$$

which is the classical weak formulation underlying finite element discretizations of the Poisson equation.

1.5.3 A Close Encounter with Finite Elements

For the Poisson problem

$$-\nabla^2 u = 1 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega,$$

with $\Omega = [-L, L]^2$, the Dirichlet Energy is

$$\Pi(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 d\Omega - \int_{\Omega} u d\Omega, \quad u \in U(\Omega)$$

Where, $U(\Omega)$ is the space of admissible functions:

$$U(\Omega) = \{u \in H^1(\Omega) : u = 0 \text{ on } \partial\Omega\}.$$

Choice of admissible basis functions. In the Ritz method, we compute an approximation $u_h(x, y)$ of the exact solution as an expansion

$$u_h(x, y) = \sum_{j=1}^N a_j \phi_j(x, y)$$

where the a_i are the unknown coefficients or *degrees of freedom* and the $\{\phi_i\}_{i=1}^N \subset U(\Omega)$ are a set of *shape functions* that must be admissible.

A crucial step in the Ritz method consists in choosing this set of *admissible basis functions*. Admissibility means first that the functions belong to the appropriate energy space, typically $H^1(\Omega)$, and that they satisfy the essential (Dirichlet) boundary conditions exactly. This requirement is not merely technical: it guarantees that the energy functional is well defined and that the minimization problem makes sense in the chosen approximation space.

Beyond admissibility, the choice of basis functions strongly influences the quality and robustness of the approximation. They must be sufficiently rich to capture the main features of the solution, while remaining simple enough to lead to tractable computations. In the context of the finite element method, this balance is achieved by using local, piecewise polynomial functions associated with a discretization of the domain, which naturally enforce boundary conditions and lead to sparse linear systems.

At this stage, it is important to emphasize that the Ritz method itself does not prescribe a unique choice of basis. Global polynomials, trigonometric functions, or problem-dependent trial functions may all be used, provided they are admissible. Finite elements can be viewed as a particularly successful and systematic way of constructing such admissible spaces, combining locality, flexibility with respect to geometry, and controlled approximation properties. We have

$$\nabla u_h = \sum_{j=1}^N a_j \nabla \phi_j, \quad |\nabla u_h|^2 = \sum_{i=1}^N \sum_{j=1}^N a_i a_j \nabla \phi_i \cdot \nabla \phi_j.$$

Plugging into Π gives a quadratic function of the coefficients:

$$\Pi(u_h) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N a_i a_j \underbrace{\int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j d\Omega}_{K_{ij}} - \sum_{i=1}^N a_i \underbrace{\int_{\Omega} \phi_i d\Omega}_{f_i}.$$

In matrix form:

$$\Pi(u_h) = \frac{1}{2} \mathbf{a}^T K \mathbf{a} - \mathbf{a}^T \mathbf{f}.$$

Stationarity $\partial \Pi / \partial a_i = 0$ for all i yields the linear system

$$K \mathbf{a} = \mathbf{f}, \quad K_{ij} = \int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j \, d\Omega, \quad f_i = \int_{\Omega} \phi_i \, d\Omega.$$

In what follows, we will use the two following admissible shape functions that vanish on $\partial\Omega$:

$$\phi_1(x, y) = (x^2 - L^2)(y^2 - L^2), \quad \phi_2(x, y) = x^2 y^2 (x^2 - L^2)(y^2 - L^2).$$

First solve with a single function. With $u_{(1)}(x, y) = a \phi_1(x, y)$, the Ritz equation is scalar:

$$a K_{11} = f_1, \quad K_{11} = \int_{\Omega} |\nabla \phi_1|^2 \, d\Omega, \quad f_1 = \int_{\Omega} \phi_1 \, d\Omega.$$

A direct polynomial integration over $\Omega = [-L, L]^2$ gives

$$K_{11} = \frac{256}{45} L^8, \quad f_1 = \frac{16}{9} L^6,$$

hence

$$a = \frac{f_1}{K_{11}} = \frac{5}{16 L^2}, \quad u_{(1)}(x, y) = \frac{5}{16 L^2} (x^2 - L^2)(y^2 - L^2).$$

Two-function solution. With $u_{(2)}(x, y) = a_1 \phi_1(x, y) + a_2 \phi_2(x, y)$, one obtains

$$K = \begin{bmatrix} \frac{256}{45} L^8 & \frac{256}{1575} L^{12} \\ \frac{256}{1575} L^{12} & \frac{2816}{33075} L^{16} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \frac{16}{9} L^6 \\ \frac{16}{225} L^{10} \end{bmatrix}, \quad K \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \mathbf{f},$$

whose solution is

$$a_1 = \frac{127}{416 L^2}, \quad a_2 = \frac{105}{416 L^6}.$$

Thus

$$u_{(2)}(x, y) = \frac{127}{416 L^2} (x^2 - L^2)(y^2 - L^2) + \frac{105}{416 L^6} x^2 y^2 (x^2 - L^2)(y^2 - L^2).$$

Comparison at $x = 0$ and $y = 0$ (the center point $(0, 0)$). At the center, $\phi_1(0, 0) = L^4$ and $\phi_2(0, 0) = 0$, hence

$$u_{(1)}(0, 0) = \frac{5}{16} L^2 \approx 0.3125 L^2, \quad u_{(2)}(0, 0) = \frac{127}{416} L^2 \approx 0.30528846 L^2.$$

For the *exact* solution (cfr. LEPL1103, your PDE introductory course) on $\Omega = [-L, L]^2$, a standard sine-series on the shifted square $(0, 2L)^2$ gives

$$u(x, y) = \sum_{\substack{m, n \geq 1 \\ m, n \text{ odd}}} \frac{64 L^2}{\pi^4} \frac{1}{mn(m^2 + n^2)} \sin\left(\frac{m\pi(x+L)}{2L}\right) \sin\left(\frac{n\pi(y+L)}{2L}\right).$$

At $(0, 0)$ this becomes (since $\sin(m\pi/2) = \pm 1$ for odd m)

$$u(0, 0) = \frac{64 L^2}{\pi^4} \sum_{\substack{m, n \geq 1 \\ m, n \text{ odd}}} \frac{\sin(m\pi/2) \sin(n\pi/2)}{mn(m^2 + n^2)} \approx 0.2946854 L^2.$$

Numerical error at the center. Therefore,

$$u_{(1)}(0,0) - u(0,0) \approx (0.3125 - 0.2946854)L^2 \approx 0.0178146 L^2,$$

$$u_{(2)}(0,0) - u(0,0) \approx (0.3052885 - 0.2946854)L^2 \approx 0.0106031 L^2,$$

so the two-function Ritz space improves the center value (about 6.0% relative error $\rightarrow$ about 3.6%).

1.5.4 Ritz sequence and energy orthogonality.

The previous example seems to indicate, at least experimentally, that enriching the approximation space leads to an improved solution. This is by no means a coincidence. In fact, this behavior is a direct consequence of the variational structure of the problem and of the way the Ritz method constructs its approximations.

In what follows, we show in a simple and transparent manner the underlying mechanism that allows the solution to systematically improve when the approximation space is enriched. The key idea is that each enrichment adds a correction *that is orthogonal, in an energy sense, to the previously explored directions*, and therefore can only decrease the value of the energy functional.

Let $\Pi : U(\Omega) \rightarrow \mathbb{R}$ be the energy functional

$$\Pi(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 d\Omega - \int_{\Omega} f u d\Omega,$$

and let u be its unique minimizer. Consider a nested sequence of finite-dimensional approximation spaces

$$V_1 \subset V_2 \subset \cdots \subset U(\Omega),$$

and define the *Ritz sequence* $\{u_n\}$ by

$$\Pi(u_n) = \min_{v \in V_n} \Pi(v).$$

Stationarity of the energy. Since u_n minimizes Π over V_n , the first variation of Π at u_n vanishes in every admissible direction $w \in V_n$:

$$\left. \frac{d}{d\varepsilon} \Pi(u_n + \varepsilon w) \right|_{\varepsilon=0} = 0 \quad \forall w \in V_n.$$

A direct computation gives

$$\int_{\Omega} \nabla u_n \cdot \nabla w d\Omega = \int_{\Omega} f w d\Omega \quad \forall w \in V_n.$$

Orthogonality between successive Ritz approximations. Because $V_n \subset V_{n+1}$, the function u_{n+1} satisfies the same stationarity condition for all test functions $w \in V_n$:

$$\int_{\Omega} \nabla u_{n+1} \cdot \nabla w d\Omega = \int_{\Omega} f w d\Omega \quad \forall w \in V_n.$$

Subtracting the two relations yields

$$\int_{\Omega} \nabla(u_{n+1} - u_n) \cdot \nabla w d\Omega = 0 \quad \forall w \in V_n.$$

In particular, choosing $w = u_n$ shows that the correction $u_{n+1} - u_n$ is orthogonal, in the energy sense, to the previous approximation.

Energy decrease along the Ritz sequence. Using the above orthogonality, one can expand the energy difference:

$$\Pi(u_{n+1}) = \Pi(u_n) - \frac{1}{2} \int_{\Omega} |\nabla(u_{n+1} - u_n)|^2 d\Omega.$$

Hence

$$\Pi(u_{n+1}) \leq \Pi(u_n),$$

with strict inequality unless $u_{n+1} = u_n$. Each enrichment of the approximation space therefore produces a genuine decrease of the energy.

Best approximation in energy. Let $v \in V_n$ be any admissible function. Using the identity

$$u - v = (u - u_n) + (u_n - v),$$

and the energy orthogonality

$$\int_{\Omega} \nabla(u - u_n) \cdot \nabla(u_n - v) d\Omega = 0,$$

one obtains

$$\int_{\Omega} |\nabla(u - v)|^2 d\Omega = \int_{\Omega} |\nabla(u - u_n)|^2 d\Omega + \int_{\Omega} |\nabla(u_n - v)|^2 d\Omega,$$

which implies

$$\int_{\Omega} |\nabla(u - u_n)|^2 d\Omega \leq \int_{\Omega} |\nabla(u - v)|^2 d\Omega.$$

Thus, among all functions in V_n , the Ritz solution u_n is the closest one to the exact solution in energy.

Geometric interpretation. Geometrically, one may think of the discrete space V_h as a curve embedded in the large ambient space $U(\Omega)$. The Ritz solution u_h is obtained by projecting the exact solution u orthogonally onto this curve, where orthogonality is understood in the sense induced by the energy functional (see Figure 1.6).

When a new admissible basis function is added, the approximation space is enriched. Geometrically, the curve becomes a surface that contains the original curve. The previous approximation space is therefore still available, but new directions are added.

As a consequence, the orthogonal projection of u onto the enriched space can only be closer to u in energy. In the worst case, the new directions are not used and the projection coincides with the one onto the original curve; in the best case, the additional directions allow the projection to move closer to the exact solution. This simple geometric picture explains why enriching the approximation space necessarily improves, or at least does not deteriorate, the Ritz approximation.

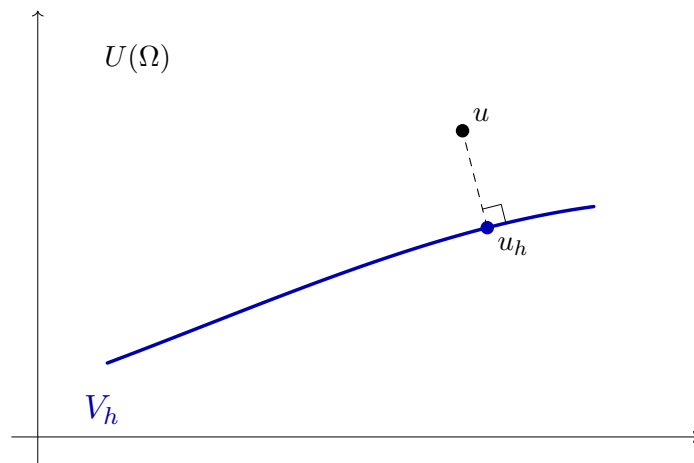


Figure 1.6: Schematic geometric picture of the Ritz method: u_h lies on the parametric manifold V_h , and u is placed along the *exact* normal direction at u_h (energy-orthogonal projection idea).

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